

Michael Lee

mhykoly@gmail.com | 714-504-3953 | whirlyfan.com | linkedin.com/in/mhykoly | github.com/WhirlyFan

Education

The University of Texas at Austin, M.S. in Computer Science, GPA: 4.0 Aug 2024 – May 2026
• **Coursework:** Natural Language Processing; Deep Learning; Android Programming; Reinforcement Learning
University of California – Irvine, B.A. in Business Economics & Intl. Studies Sep 2018 – Jun 2022

Technologies

Languages: Python, JavaScript, TypeScript, HTML/CSS, SQL

Technologies: React, Next.js, Vite, Redux, Express, Apollo, GraphQL, Django, Flask, Git, PostgreSQL, PyTorch, Kotlin, Supabase, Docker, SimPy, Streamlit, Hatchet, Porter, Redis, Claude Code

Experience

Full Stack Software Engineer, Usul – San Francisco, CA Jan 2026 – Present

- Built AI-powered search using an MCP server to query contract data, alongside keyword search, enabling clients to discover matching government contracts by capabilities
- Led international feature rollout and onboarded clients globally across multiple regions
- Designed frontend overhaul, cutting load times by 50% via cache invalidation and real-time cross-client updates using server-sent events
- Designed backend architecture to scale from seed to enterprise. Includes implementing CI/CD workflows, migration management, ORMs, backend service layers, and AI orchestration
- Engineered LLM workflows to automatically process and surface relevant contract recommendations to customers
- Leveraged AI-assisted development workflows to work across multiple tickets simultaneously, increasing throughput

AI and Data Science Intern, Nouryon – Houston, TX Jun 2025 – Aug 2025

- Designed a digital simulation for a multinational chemical manufacturing product line using years of historical data
- Identified process bottlenecks to recommend cost-saving improvements

Full Stack Software Engineer, Apero Health (YC Startup) – San Francisco, CA May 2023 – Jul 2023

- Designed a dynamic forms builder enabling healthcare clients to generate HIPAA-compliant forms
- Reduced legacy code duplication by 80% by overhauling the frontend to use a centralized component library

Projects

Lister github.com/WhirlyFan/Lister

- Full-stack MyAnimeList clone with live chat, followers, anime lists, and reviews using React/Redux, Flask, and SQLAlchemy
- Implemented real-time communication via WebSockets and integrated a third-party API for detailed anime data

Breezy Apr 2025

- Built an Android application in Kotlin modeled after Instagram, supporting CRUD operations on posts, comments, likes, and messages
- Backend utilizes Supabase for authentication, real-time messaging, and PostgreSQL triggers and functions

Research

Enhancing Natural Language Inference Robustness Through Adversarial Dataset Fine-Tuning Dec 2024

- Fine-tuned an Electra-small model with adversarial datasets, achieving an 8.09% accuracy increase on out-of-distribution data, demonstrating improved robustness and generalization